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1. Project Technical Feasibility

- **Definition**: Assesses whether the project can be executed with the available technology, tools, and technical expertise.
- **Example**: In designing a smart irrigation system, checking if the sensors and microcontrollers (like Arduino or Raspberry Pi) can handle soil moisture detection and automated water flow.
- **Use in Research**: Ensures the proposed design is technically achievable and not beyond current capabilities.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the technical feasibility, operational feasibility, testing, evaluation results, analysis, deployment results, project economic feasibility, and schedule feasibility. The findings of the evaluation suggest that the device should be improved further.

Project Technical Feasibility

Project Design Technical Discussion

The project's developers created a device capable of efficiently drying and pressing pandan leaves to help the Handicraft Workers of Barangay Zone VI, Luisiana, Laguna, to improve productivity using the help of technology. The device is designed with a control box that includes several components to ensure optimal performance of the device. The temperature is controlled by a PID controller that maintains the desired output by adjusting the power supplied at 12V to the heating elements based on real-time sensor feedback. Also, the drying process of the pandan leaves is displayed through an LCD screen that can show the current status of the drying and pressing, the elapsed time, and the device's current temperature. The device has eight (8) infrared heating elements inside that provide heat distribution to effectively reduce pandan leaves' moisture to avoid leaf damage. Four (4) fans operate within the chamber to circulate hot air to make sure to ensure consistent temperature of the device and uniform drying.

After the drying process, the device enters pressing mode, where the leaves are pressed between the roller system that is driven by a wiper motor with

Software Specification

The Pandaiku: Drying and Pressing System software is compatible with Arduino IDE, allowing for simulation. In the project, programmers used C++ as the programming language. It facilitates writing, compiling, and uploading code onto the Arduino module. When the device is released at a specific time and temperature, the software running on Arduino Mega executes command functions to control the devices. It controls the activation of the infrared heaters, the timing sequences for the roller motor, and the real-time temperature monitoring. Upon reaching a pre-set temperature, the PID controller, pre-configured through the IDE, regulates it via on-off control. Limit switch signals are detected through digital input detection to activate auto-stop functions for the roller motor.

Programming Environment

The developers employed the Arduino IDE as the programming environment to design and implement the control system of drying and pressing device for pandan leaves. The Arduino MEGA is programmed to handle various hardware components, including the PID controller, solid-state relay, channel relay, and step-down voltage regulator. The device also used ceramic heating element, thermocouple to monitor the temperature inside the chamber, exhaust fan to regulate the air inside, and wiper motor for the pressing mechanism of the device. Device also used a 20x4 LCD display, and keypad, allowing the user to input specific time and real-time monitoring. The buzzer can give awareness to the operator when the operation is done, on the other hand the pin headers make stable connections to all components used in the device. The program ensures the coordinated operation to achieve enough drying and pressing of pandan leaves.

Hardware Specification

The Arduino Mega 2560 is a microcontroller board based on the ATmega2560. It has 54 digital input/output pins (of which 15 can be used as PWM outputs), 16 analog inputs, 4 UARTs (hardware serial ports), a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a



https://i.ebayimg.com/images/g/fs4AAOSwNiJi7B2k/s-l1600.jpg

Figure 4.1 Arduino Mega



https://www.fabtolab.com/image/cache/data/Power/Controls/KY-019-500x500.jpg

Figure 4.8. 1 Channel Relay

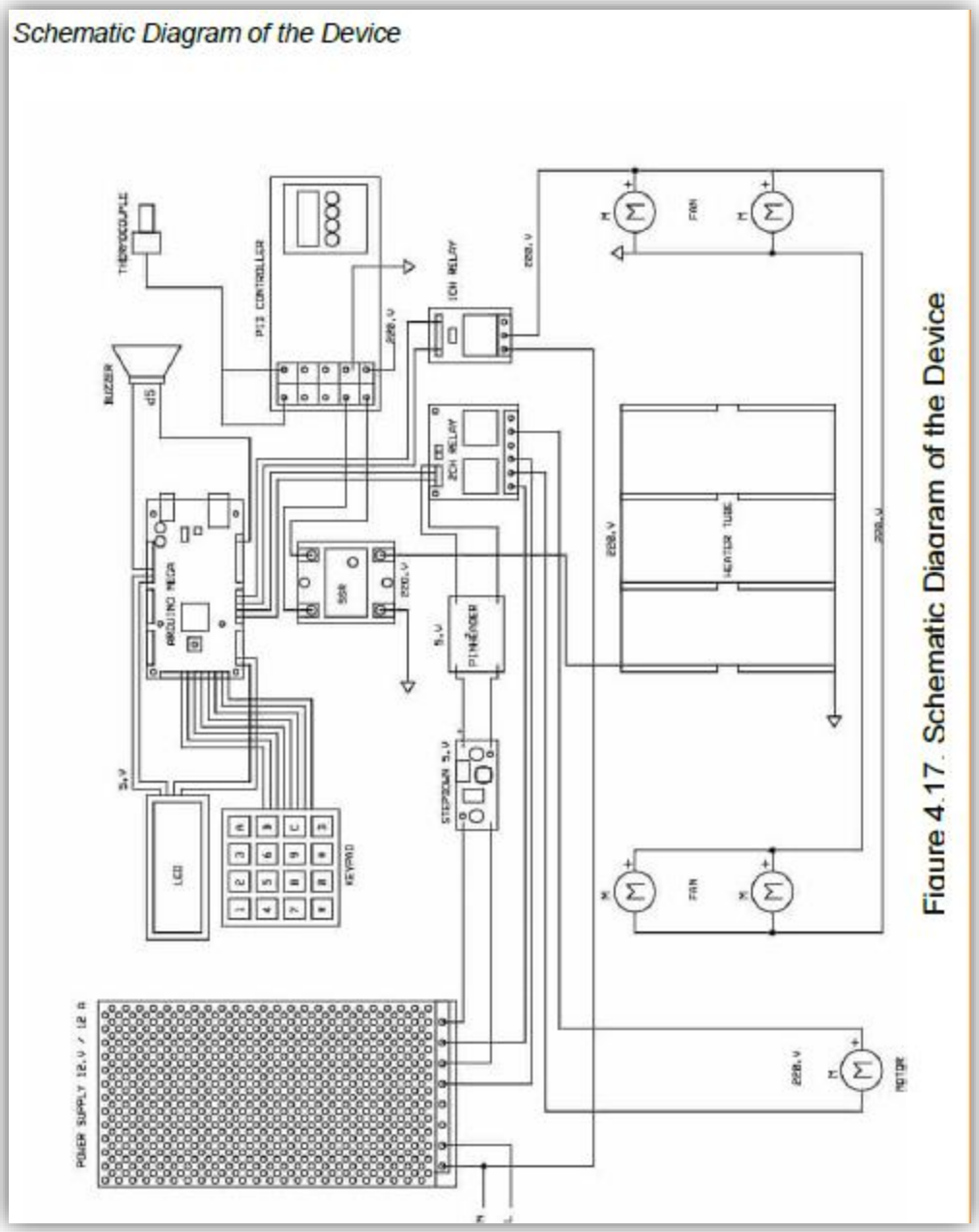


Figure 4.17. Schematic Diagram of the Device

2. Project Operational Feasibility

- **Definition**: Examines if the proposed system will work smoothly in real-world operations and if users can easily adopt it.
- **Example**: Testing if farmers can easily operate the mobile app for the irrigation system without advanced technical knowledge.
- **Use in Research**: Confirms that the system is user-friendly and sustainable in daily use.

Project Operational Feasibility

Design of Project

Based on the figure 4.11 it shows the actual image of the Pandaiku, an Arduino based device that can be used for drying and pressing the pandan leaves that usually tend to made a handicraft product. It is constructed from stainless steel because of the its advantages such as flexibility and load-bearing capabilities which ensure that the components are protected and secured. This thoughtful design enhances the device's overall performance and durability, making it a practical solution for long-term use.



Figure 4.18. Pandaiku Drying and Pressing



Figure 4.19. Liquid Crystal Display



Figure 4.20 Heating and Pressing inside the Device

3. Testing, Evaluation Results, and Analysis

- **Definition**: Process of verifying and validating the design against requirements, then analyzing the outcomes.
- **Example**: Running performance tests on the irrigation system to measure water efficiency, then analyzing data to compare with traditional irrigation methods.
- **Use in Research**: Provides evidence that the project meets its objectives and identifies strengths/weaknesses.

Testing, Evaluation Results, and Analysis

Program Testing Results

Below is the figure of the actual testing results on software functionalities of Pandaiku made by the developers to ensure that the device executes its functions smoothly and was implemented in the Arduino successfully. This series of testing done by the developers is also important to avoid errors and fix bugs immediately to resolve issues regarding the device.

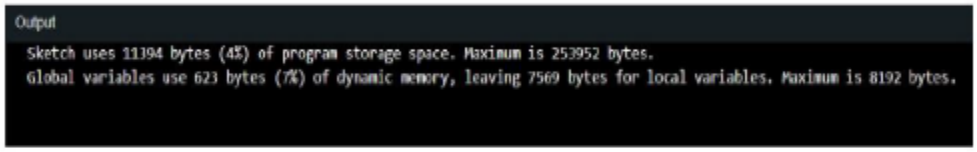


Figure 4.21. Program Testing Results

Hardware Testing

Table 4.1. Hardware Testing Results of LCD Screen

Day	Functionality	Reliability	Accuracy
1	/	X	/
2	/	X	/
3	/	/	/
4	/	/	/
5	/	/	/

Runs Smoothly (/) Error (X)

Table 4._ shows the testing made by the developers on the LCD screen to test its functionalities, reliability, and accuracy. Upon the series of testing made by the developers on 5 consecutive days, the LCD screen executes its functions

Table 4._. Hardware Testing Results of LCD Screen

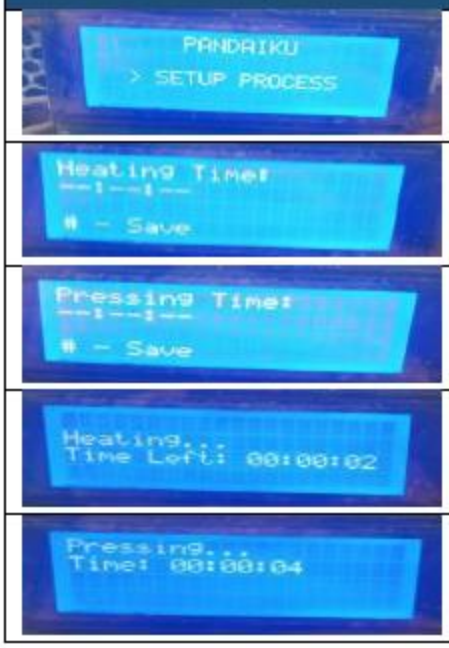
Actual Photo of LCD during Testing	Analysis
	The LCD screen functioned accordingly to what it is expected to do or its purpose without any delays or errors.

Table 4.7. Drying 50-70 Pieces of Pandan Leaves at 80°C

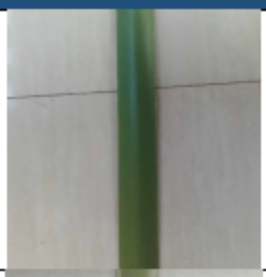
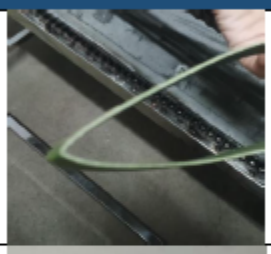
Actual Photo of Pandan Leaves	Time (minutes)	Temperature (°C)	Analysis
	0	65	Pandan leaves is fresh and is color green
	5	70	Pandan leaves starts to dry at the edges.

Table 4.8. Pressing Pandan Leaves Testing Results

Actual Photo of Pandan Leaves	Time (minutes)	Analysis
	0	Pandan leaves is brittle.
	5	Pandan leaves breaks or cracks easily.

Summary of Testing

Below is the summary of all the testing that the developer conducted to ensure that the device meet the expected outcome and satisfaction of the user. The temperature monitoring and regulation functioned successfully through the use of PID controller with thermocouple sensor to read real-time data which is the temperature inside the device. Regarding to the drying and pressing testing of 50

Table 4.10. Summary of Specific Objective Testing

Testing	Expected Outcome	Actual Outcome
Temperature Monitoring and Regulation Testing	The device will be able to maintain and monitor the temperature inside using the PID controller.	The device was able to maintain and monitor the temperature inside using the PID controller.
Drying 50-70 pieces of Pandan Leaves	The device will be able to dry 50-70 pieces of pandan leaves with the appropriate amount of dryness depending on the set temperature	The device was able to dry 50-70 pieces of pandan leaves with the appropriate amount of dryness depending on the set temperature
Pressing Testing	The device will be able to press pandan leaves with the right level of pliability.	The device was able to press pandan leaves with the right level of pliability.
Notification Testing	The buzzer will be able to produce sounds once the pressing process is complete.	The buzzer was able to produce sounds once the pressing process is complete.

Evaluation results

Pandaiku was evaluated by five (5) CpE practitioners, five (5) members of Luisiana Weavers Organization as well as one (1) Agriculturist from their municipality. This is to test and assess whether Pandaiku aligns and met the international standards and requirements regarding systems and devices. The evaluation is based on ISO/IEC 25010 which is composed of two parts. The first part of the evaluation is about the software standards which includes functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. On the other hand, part two is about hardware evaluation which asses the components, technicalities, as well as the structure of the device. The evaluation was conducted on a face-to-face modality where they can physically see and test the device, provide feedback and suggestions for further improvements.

CpE Practitioners' Evaluation Results

Table 4.11. CpE Practitioners' Evaluation Results Based on Functional Suitability

Criteria	Mean	Verbal Interpretation
Functional Completeness	4.8	Highly Acceptable
Functional Correctness	4.4	Very Acceptable
Functional Appropriateness	4.8	Highly Acceptable
Overall Mean	**4.67**	**Highly Acceptable**

Figure 4.11 shows the result of five (5) CpE practitoners under the category of functional suitability which checks if the product has all the necessary functions

Table 4.12. CpE Practitioners' Evaluation Results Based on Performance Efficiency

Criteria	Mean	Verbal Interpretation
Time Behavior	4.6	Highly Acceptable
Resource Utilization	4.8	Highly Acceptable
Capacity	4.6	Highly Acceptable
Overall Mean	**4.67**	**Highly Acceptable**

Table 4.13. CpE Practitioners' Evaluation Results Based on Compatibility

Criteria	Mean	Verbal Interpretation
Co-existence	4.6	Highly Acceptable
Interoperability	4.6	Highly Acceptable
Overall Mean	**4.6**	**Highly Acceptable**

Table 4.14. CpE Practitioners' Evaluation Results Based on Usability

Criteria	Mean	Verbal Interpretation
Appropriateness Recognizability	5	Highly Acceptable
Learnability	4.6	Highly Acceptable
Operability	4.6	Highly Acceptable
User Error Protection	4.4	Very Acceptable
User Interface Aesthetic Accessibility	4.8	Highly Acceptable
Overall Mean	**4.68**	**Highly Acceptable**

Reliability, Security, Maintainability, Portability,

Table 4.19. CpE Practitioners' Hardware Evaluation Results

Criteria	Mean	Verbal Interpretation
Q1	4.6	Highly Acceptable
Q2	4.2	Very Acceptable
Q3	4.8	Highly Acceptable
Q4	4.8	Highly Acceptable
Q5	4.8	Highly Acceptable
Overall Mean	**4.64**	**Highly Acceptable**

4. Project Benefits

- **Definition**: The advantages or value the project will bring to stakeholders (economic, social, environmental, or technical).
- **Example**: The irrigation system reduces water consumption by 30%, saves costs, and improves crop yield.
- **Use in Research**: Demonstrates the project's contribution and justifies why it should be implemented.

Project Benefits

The benefit of this project is to help the weavers to ease their manual labor in pandan leaf processing specially in drying and pressing of pandan leaves. This device will reduce the consumed time required for the two processes and help the weavers to produce more products which will lead them to have an additional income.

Table 4.31. Comparison Between Traditional and Arduino-Based Methods for Pressing and Drying Pandan Leaves

Without Arduino-based Pandaiku	With Arduino-based Pandaiku
Taking too long to dry and press the pandan leaves	Timeless and Easy Method for Pressing and Drying the pandan leaves
The pandan weavers only know the old traditional method of drying the pandan leaves by the use of sun drying which it is placed into the open field	The pandan weavers learns a new method of drying with pressing of pandan leaves in possible shortest time
In pressing the pandan leaves weavers tend to use their traditional method which is to use '*ilohan*' which is a large	In pressing the pandan leaves it does not require too much labor which

5. Project Economic Feasibility

- **Definition**: Assesses whether the project is financially viable by comparing costs versus benefits.
- **Example**: Calculating if the cost of sensors, controllers, and installation can be recovered by the savings in water bills and increased harvest revenue within 2 years.
- **Use in Research**: Ensures stakeholders that the project is cost-effective.

Project Economic Feasibility

Table 4.33.Total Budget for the Project

BILLS OF MATERIALS		
PARTS NAME	**QUANTITY**	**PRICE**
Caster Wheel W/ Lock	4	₱640.00
Lock	4	₱200,00
Sprocket 40x20	2	₱700.00
Sprocket 40x15	2	₱600.00
Chain #40	4	₱920.00
Wiper Motor	1	₱1,750.00
Limit Switch	2	₱100.00
Heater	8	₱15,200.00
Blower Fan 220v	4	₱2,400.00
6" Tube	1	₱7,000.00
Plainsheet 202	5	₱9,500.00
Piano Hinges	2	₱100.00
Tubular 1/2 X 1/2	7	₱1,610.00
Tubular 1/2 X 1	1	₱350.00
RKC	1	₱2,200.00
12v 12.5a Switching Power Supply 150w	1	₱849.00
Lcd 20x4 Dispaly I2c White On Blue	1	₱278.00
Buzzer Continuous Sound Electronic Active Buzzer Sfm-27 Dc 3v-24v	1	₱50.00
Arduino Mega 2560 R3 Board Based On Arduino	1	₱1,049.00
4x4 Matrix Membrane Keypad	1	₱50.00
Dc-Dc Buck Converter LM2596S	1	₱60.00
Labor	3	₱10,000.00
	TOTAL	₱59,406

Operational Cost Analysis

The operational cost analysis for drying pandan leaves involves assessing the expenses related to electricity usage. During a test conducted on April 2025, the drying and pressing process of the device reached a maximum power consumption of 0.684 kWh. As of that date, the electricity rate was ₱13.00 kWh. In a typical operation, drying of 70 pieces of pandan leaves consumes approximately power over 30 minutes and the pressing is for 15 minutes, the total operational cost per cycle for drying and pressing of pandan leaves is approximately ₱9.00.

Potential Profitability Analysis

A sample profitability calculation was performed for drying and pressing 70 leaves of pandan leaves. In one process of it contains 30 minutes drying + 15 minutes pressing which cost the total operational electricity of ₱9.00.

To produce one basket its average that it needed 30 pandan leaves so the 70 leaves can approximately make 2 baskets (70 ÷ 30 ≈ 2.33, rounded down to 2 for practical output). Given a market price of ₱200.00 per baskets, the total gross revenue per cycle is ₱400 (2 baskets x ₱200.00). Subtracting the operational cost of ₱9.00 result in a net profit per cycle is ₱191.00 (₱200.00-₱9.00).

Assume for one day cycles, this leads to a daily profit of ₱18,768.00. Over a 30-day month, the potential monthly profit would be ₱563,040.00, showing that the device has a strong income potential foe pandan based handicraft production.

6. Project Schedule Feasibility

- **Definition**: Determines if the project can be completed within the planned timeline.
- **Example**: Checking if the irrigation system prototype can be built, tested, and deployed within 6 months.
- **Use in Research**: Aligns project goals with realistic timeframes to avoid delays and resource waste.

Project Schedule Feasibility

Table 4.34. Project Development Schedule

Description	Activity	Date Started	Date Finished	Duration
Plan	User Requirements	02-17-2025	02-28-2025	14 days
	System Requirements	02-28-2025	03-16-2025	16 days
	Hardware Requirements	02-28-2025	03-16-2025	16 days
Design	Design of the Prototype	01-16-2025	03-20-2025	34 days
	System Design	02-09-2025	03-09-2025	31 days
Develop	Components Gathering	02-02-2925	02-16-2025	14 days
	Building and Wiring	02-27-2025	03-31-2025	31 days